DBMS Week 5 – Payment Transaction Management SystemScenario

The Travel Booking Management System allows customers to book travel packages and make payments for their bookings. The system maintains payment transaction details, including payment date, payment mode, transaction amount, and payment status. It tracks successful, failed, and pending transactions and generates reports to help monitor payments and revenue.

Create Payment Table:

Store Payment Details:

```
SQL> connect
Enter user-name: C##TRAVEL_BOOKING
Enter password:
Connected.
SQL> select*from customer;

CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                               PHONE CREATED_A
------------------------------ ---------- ---------
C001                 Arun                 Kumar
arun@gmail.com                 9876543210 10-JAN-26

C002                 Priya                Sharma
priya@gmail.com                9876543211 15-JAN-26

C003                 Rahul                Raj
rahul@gmail.com                9876543212 05-FEB-26


CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                               PHONE CREATED_A
------------------------------ ---------- ---------
C004                 Divya                Suresh
divya@gmail.com                9876543213 12-FEB-26

C005                 Karthik              Mohan
karthik@gmail.com              9123456780 01-MAR-26

C006                 Sneha                Ravi
sneha@gmail.com                9876543215 08-MAR-26


CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                               PHONE CREATED_A
------------------------------ ---------- ---------
C007                 Vishnu               Kumar
vishnu@gmail.com               9876543216 20-MAR-26

C008                 Ananya               Krishnan
ananya@gmail.com               9876543217 02-APR-26
```

Manage Successful and Failed Transactions

Tasks

Write SQL queries to:

1. Display all successful payments.
2. 2. Find failed transactions.
3. 3. Count total successful and failed payments.

```
SQL> SELECT *
  2  FROM Payment
  3  WHERE Payment_Status = 'Success';

PAYMENT_ID BOOKING_ID PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
P1         B1         10-SEP-26 UPI                  Success
             24000

P2         B2         11-SEP-26 Card                 Success
             36000

P4         B4         13-SEP-26 UPI                  Success
             65000


PAYMENT_ID BOOKING_ID PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
P5         B5         14-SEP-26 Card                 Success
             25000

P7         B7         16-SEP-26 UPI                  Success
             45000
```

```
SQL> SELECT *
  2  FROM Payment
  3  WHERE Payment_Status = 'Failed';

PAYMENT_ID BOOKING_ID PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
P8         B8         17-SEP-26 UPI                  Failed
             55000
```

```
SQL> SELECT Payment_Status,
  2         COUNT(*) AS Total_Payments
  3  FROM Payment
  4  WHERE Payment_Status IN ('Success', 'Failed')
  5  GROUP BY Payment_Status;

PAYMENT_STATUS       TOTAL_PAYMENTS
-------------------- --------------
Failed                            1
Success                           5
```

Analyze Payment Methods Used by Customers

Generate payment analysis reports

Report 1 : Payment Mode Analysis

Find:

- Number of transactions using UPI
- Number of card payments
- Most preferred payment method

```
SQL> SELECT COUNT(*) AS Total_UPI_Payments
  2  FROM Payment
  3  WHERE Payment_Mode = 'UPI';

TOTAL_UPI_PAYMENTS
------------------
                 4
```

```
SQL> SELECT COUNT(*) AS Total_Card_Payments
  2  FROM Payment
  3  WHERE Payment_Mode = 'Card';

TOTAL_CARD_PAYMENTS
-------------------
                  2
```

```
SQL> SELECT Payment_Mode,
  2         COUNT(*) AS Total_Transactions
  3  FROM Payment
  4  GROUP BY Payment_Mode
  5  ORDER BY Total_Transactions DESC;

PAYMENT_MODE         TOTAL_TRANSACTIONS
-------------------- ------------------
UPI                                   4
Card                                  2
Cash                                  1
Net Banking                           1
```

Report 2: Revenue Analysis

Find:

- Total revenue generated
- Revenue by payment method
- Average transaction amount

```
SQL> SELECT COUNT(*) AS Total_Debit_Card_Payments
  2  FROM Payment
  3  WHERE Payment_Mode = 'Debit Card';

TOTAL_DEBIT_CARD_PAYMENTS
-------------------------
                        0
```

```
SQL> SELECT SUM(Transaction_Amount) AS Total_Revenue
  2  FROM Payment
  3  WHERE Payment_Status = 'Success';

TOTAL_REVENUE
-------------
       195000
```

```
SQL> SELECT Payment_Mode,
  2         SUM(Transaction_Amount) AS Total_Revenue
  3  FROM Payment
  4  WHERE Payment_Status = 'Success'
  5  GROUP BY Payment_Mode
  6  ORDER BY Total_Revenue DESC;

PAYMENT_MODE         TOTAL_REVENUE
-------------------- -------------
UPI                         134000
Card                         61000
```

```
SQL> SELECT AVG(Transaction_Amount) AS Average_Transaction_Amount
  2  FROM Payment
  3  WHERE Payment_Status = 'Success';

AVERAGE_TRANSACTION_AMOUNT
--------------------------
                     39000
```

Report #: Customer Payment History

Display:

- Customer name
- Order ID
- Payment mode
- Amount
- Payment status

```
SQL> SELECT
  2      C.FIRST_NAME,
  3      C.LAST_NAME,
  4      B.BOOKING_ID,
  5      P.PAYMENT_MODE,
  6      P.TRANSACTION_AMOUNT,
  7      P.PAYMENT_STATUS
  8  FROM CUSTOMER C
  9  INNER JOIN BOOKING B
 10      ON C.CUSTOMER_ID = B.CUSTOMER_ID
 11  INNER JOIN PAYMENT P
 12      ON B.BOOKING_ID = P.BOOKING_ID;

FIRST_NAME           LAST_NAME            BOOKING_ID PAYMENT_MODE
-------------------- -------------------- ---------- --------------------
TRANSACTION_AMOUNT PAYMENT_STATUS
------------------ --------------------
Arun                 Kumar                B1         UPI
             24000 Success

Priya                Sharma               B2         Card
             36000 Success

Rahul                Raj                  B3         Net Banking
             55000 Pending


FIRST_NAME           LAST_NAME            BOOKING_ID PAYMENT_MODE
-------------------- -------------------- ---------- --------------------
TRANSACTION_AMOUNT PAYMENT_STATUS
------------------ --------------------
Divya                Suresh               B4         UPI
             65000 Success

Karthik              Mohan                B5         Card
             25000 Success

Sneha                Ravi                 B6         Cash
             30000 Pending


FIRST_NAME           LAST_NAME            BOOKING_ID PAYMENT_MODE
-------------------- -------------------- ---------- --------------------
TRANSACTION_AMOUNT PAYMENT_STATUS
------------------ --------------------
Vishnu               Kumar                B7         UPI
             45000 Success

Ananya               Krishnan             B8         UPI
             55000 Failed


8 rows selected.
```

Payment Analysis

```
SQL> SELECT Payment_Mode, COUNT(*) AS Total_Transactions
  2  FROM Payment
  3  GROUP BY Payment_Mode
  4  ORDER BY Total_Transactions DESC;

PAYMENT_MODE         TOTAL_TRANSACTIONS
-------------------- ------------------
UPI                                   4
Card                                  2
Cash                                  1
Net Banking                           1

SQL>
```